# 10 Divisibility and Factorisation in Rings

In this section all rings are **commutative** with an identity element $1 \neq 0$.

## 10.1 Euclidean Domains

Let $R$ be an integral domain (ring without zero divisors). A function

$$N : R \setminus \{0_R\} \to \mathbb{N}$$

is called a **norm**.

A norm $N$ is called multiplicative if :

$$N(ab) = N(a)N(b)$$

for all $a, b \in R \setminus \{0_R\}$.

$R$ is called an **Euclidean domain (ED)** if there exists a norm $N$ on $R$ such that for all $a, b \in R$ with $b \neq 0$ there exists $q, r \in R$ so that

$$a = bq + r$$

with $r = 0$, or $N(r) < N(b)$ [$r$ is the remainder of the division of $a$ by $b$].

**Example 10.1.**
**(1)** $\mathbb{Z}$ is an Euclidean domain with multiplicative norm

$$N : \mathbb{Z} \setminus \{0\} \to \mathbb{N}$$

$$a \mapsto |a|.$$

**(2)** $\mathbb{Q}[X]$, or more generally $F[X]$ where $F$ is a field, is an Euclidean domain with norm

$$N : \mathbb{Q}[X] \setminus \{0\} \to \mathbb{N}$$

$$f(X) \mapsto \deg(f).$$

**(3)** $\mathbb{Z}[i] = \{a + ib : a, b \in \mathbb{Z}\}$ is an Euclidean domain with norm

$$N : \mathbb{Z}[i] \setminus \{0\} \to \mathbb{N}$$

$$a + ib \ \mapsto \ N(a + ib) = a^2 + b^2.$$

First we show $N$ is a multiplicative norm. Let $a + ib, c + id \in \mathbb{Z}[i]$ then

$$N[(a + ib)(c + id)] = N[(ac - bd) + i(ad + bc)] = (ac - bd)^2 + (ad + bc)^2$$
$$= (ac)^2 - 2acbd + (bd)^2 + (ad)^2 + 2adbc + (bc)^2 = (ac)^2 + (bd)^2 + (ad)^2 + (bc)^2$$
$$a^2c^2 + b^2d^2 + a^2d^2 + b^2c^2 = (a^2 + b^2)(c^2 + d^2) = N(a + ib)N(c + id).$$

Next we show the Euclidean division property. Let $\alpha = a + ib$, and $0 \neq \beta = c + id \in \mathbb{Z}[i]$. Then

$$\frac{\alpha}{\beta} = \frac{(a + ib)(c - id)}{c^2 + d^2} = \frac{ac + bd}{c^2 + d^2} + i\frac{bc - ad}{c^2 + d^2} \in \mathbb{C}.$$

Write

$$r = \frac{ac+bd}{c^2+d^2} \in \mathbb{Q} \qquad s = \frac{bc-ad}{c^2+d^2} \in \mathbb{Q}.$$

Let $p, q \in \mathbb{Z}$ such that $|r-p| \leq \frac{1}{2}$, $|s-q| \leq \frac{1}{2}$, and let

$$\theta = \frac{\alpha}{\beta} - (p+iq) = (r+is) - (p+iq) = r - p + i(s-q) \in \mathbb{Q}(i)$$

and

$$\gamma = \beta\theta = \alpha - (p+iq)\beta \in \mathbb{Z}[i].$$

Thus

$$\alpha = (p+iq)\beta + \gamma$$

with $\gamma \in \mathbb{Z}[i]$.

We show $N(\gamma) < N(\beta)$. We have $N(\gamma) = N(\beta)N(\theta)$ and

$$N(\theta) = (r-p)^2 + (s-q)^2 \leq \frac{1}{4} + \frac{1}{4} = \frac{1}{2}$$

which implies $N(\gamma) \leq \frac{N(\beta)}{2} < N(\beta)$ as required.

A **principal ideal domain (PID)** is an integral domain $R$ with the property that every ideal of $R$ is principal generated by one element: if $I \subset R$ is an ideal then

$$I = (a)_R = \{ab : b \in R\}$$

for some element $a$ of $R$.

**Theorem 10.1.**
*Let $R$ be an Euclidean domain then $R$ is a principal ideal domain (PID)*

$$\textbf{ED} \implies \textbf{PID}.$$

*Proof.* Let $\{0_R\} \neq I$ be an ideal of $R$, we show it is principal. Let

$$n = \min\{N(\tilde{a}) : \tilde{a} \in I, \tilde{a} \neq 0_R\},$$

and $a \in I$ with $N(a) = n$. We show $I = (a)_R$.

Clearly $(a)_R \subset I$ since $a \in I$, we show $I \subset (a)_R$.

Let $b \in I$ then $b = \alpha a + r$ with either $r = 0$ or $N(r) < N(a)$. Also $r = b - \alpha a \in I$ since $a, b \in I$, $\alpha a \in I$ since $I$ is an ideal, and $(I, +)$ is a subgroup of $(R, +)$. This implies $r = 0$ necessarily by the minimality of $a$ [a non zero $r \in I$ with $N(r) < N(a)$ would contradict the minimality of $a$] hence $b = \alpha a \in (a)_R$ and $I \subset (a)_R$. □

**Example 10.2.**
**(1)** $\mathbb{Z}$, $\mathbb{Q}[X]$ (or more generally $F[X]$ where $F$ is a field), and $\mathbb{Z}[i]$ are PID since they are Euclidean domains.

**(2)** $R = \mathbb{Z}[\sqrt{-5}] = \{a + b\sqrt{-5} : a, b \in \mathbb{Z}\}$ is NOT a PID, in particular $R$ is NOT an ED. Note that

$$N : \mathbb{Z}[\sqrt{-5}] \to \mathbb{N}$$

$$a+b\sqrt{-5} \ \mapsto \ N(a+b\sqrt{-5}) = a^2+5b^2$$

is a multiplicative norm (check this fact). Further $N(a+b\sqrt{-5}) = a^2+5b^2 = 1 \iff a^2 = 1, b = 0$ hence $a+ib \in \{1,-1\}$.

The ideal

$$I = (3, 2+\sqrt{-5})_R = \{3x + (2+\sqrt{-5})y : x, y \in R\}$$

is not principal. Indeed suppose the contrary $I = (c+d\sqrt{-5})_R$ is principal. Then $3 = \alpha(c+d\sqrt{-5})$ $(*)$, and $2+\sqrt{-5} = \beta(c+d\sqrt{-5})$ $(**)$, for some $\alpha, \beta \in R$. Now $(*)$ implies $N(3) = 9 = N(\alpha)N(c+d\sqrt{-5}) = N(\alpha)(c^2+5d^2)$ hence $c^2+5d^2$ equals 1, 3, or 9. We have the following possibilities:

$N(c+d\sqrt{-5}) = 9 \Longrightarrow N(\alpha) = 1 \Longrightarrow \alpha \in \{-1,1\} \Longrightarrow c+d\sqrt{-5} \in \{-3,3\}$ which is impossible by $(**)$ since $\beta$ would be 1 or $-1$.

$N(c+\sqrt{-5}d) = 3 = c^2+5d^2$ which is not possible.

$N(c+\sqrt{-5}d) = 1 \Longrightarrow c+d\sqrt{-5} \in \{-1,1\} \Longrightarrow I = R$ which is not possible. Indeed if $1 = 3x+(2+\sqrt{-5})y$ for some $x, y \in R$ then $2-\sqrt{-5} = 3(2-\sqrt{-5})x+(2+\sqrt{-5})(2-\sqrt{-5})y = 3(2-\sqrt{-5})x+9y$ which implies 3 divides $2-\sqrt{-5}$: $2-\sqrt{-5} = 3v \Longrightarrow N(v) = 1$ hence $v \in \{1,-1\}$ which is not possible.

**(3)** $\mathbb{Z}[X]$ is NOT a PID since the ideal $(2, X)$ is not principal. In particular $\mathbb{Z}[X]$ is NOT an ED.

**Definition 10.1.**
Let $R$ be a ring and $a, b \in R$. **(1)** We say that $b$ *divides* $a$ (or $a$ is a *multiple* of $b$), and write $b/a$, if $a = bc$ for some $c \in R$. **(2)** A *greatest common divisor* of $a$ and $b$ is an element $d \in R$ such that:

**(i)** $d/a$ and $d/b$.

**(ii)** If $d' \in R$, $d'/a$ and $d'/b$ then $d'/d$.

In this case we write $d = \gcd(a, b)$.

**Remark 10.1.**
$\gcd(a, b)$ is not unique in general, if $d = \gcd(a, b)$ then $ud = \gcd(a, b)$ for any unit $u \in R^\times$ (check this fact).

**Proposition 10.2.**
*Let $R$ be a PID and $a, b \in R$. Then* $\gcd(a, b)$ *exists and*

$$(a)_R + (b)_R = (d)_R$$

*where* $d = \gcd(a, b)$*. Further,* $d = \gcd(a, b)$ *is unique up to multiplication by a unit, and we have a Bézout identity:*

$$d = sa + rb$$

*for some $s, r \in R$.*

*Proof.* The ideal $I = (a)_R + (b)_R$ is principal hence $(a)_R + (b)_R = (d)_R$ for some $d \in R$, we show $d = \gcd(a, b)$. We have $a = 1a+0_R b \in (d)_R \Longrightarrow a = \alpha d$ for some $\alpha \in R$ which implies $d/a$. Similarly $b = 1b+0_R a \in (d)_R \Longrightarrow b = \beta d$ for some $\beta \in R$ which implies $d/b$.

Also $d = d1_R \in (d)_R = (a)_R + (b)_R$, hence

$$d = ra + sb \qquad (*)$$

for some $s, r \in R$.

We show $d = \gcd(a, b)$. Assume $d' \in R$, $d'/a \iff a = d'x$, and $d'/b \iff b = d'y$ for some $x, y \in R$. The relation $(*)$ above implies $d = sd'x + rd'y = d'(sx + ry)$ and $d'/d$ which shows $d = \gcd(a, b)$. □

**Remark 10.2.**
In an Euclidean domain $R$ the $\gcd(a, b)$ exists for all $a, b \in R$ and can be computed by the Euclidean algorithm (as for integers):

$$a = sb + r \qquad N(r) < N(b)$$
$$b = s_1 r + r_1 \qquad N(r_1) < N(r)$$

. . . . . .

$$r_{k-2} = s_k r_{k-1} + r_k \qquad N(r_k) < N(r_{k-1})$$
$$r_{k-1} = s_{k+1} r_k$$

This algorithm has to stop after finitely many steps since $0 \leq N(r_k) < N(r_{k-1}) < \ldots < N(r_1) < N(r)$, and $\gcd(a, b) = r_k$: the last nonzero remainder in the above algorithm.

**Proposition 10.3.**
*Every nonzero prime ideal in a PID is a maximal ideal.*

*Proof.* Let $R$ be a PID and $P = (p)_R \subset R$ a nonzero prime ideal. Hence $p \neq 0_R$ for otherwise $P = \{0_R\}$ would be the zero ideal. By Proposition 9.1 there exists a maximal ideal $M \subsetneq R$ containing $P$. We show $M = P$. Since $R$ is a PID the ideal $M = (m)_R$ must be principal say generated by $m \in R$. Now $p = p1_R \in P \subset M = (m)_R$ hence $p = mx$ for some $x \in R$. Since $P$ is a prime ideal and $p = mx \in P$ either $m \in P$ or $x \in P$. If $m \in P$ then $M = (m)_R \subset P$, hence $M = (m)_R = P = (p)_R$ and the proof is complete. If $x \in P$ write $x = py$ for some $y \in R$. Thus $p = xm = pym \Longrightarrow p(1 - ym) = 0 \Longrightarrow 1 - ym = 0 \Longrightarrow ym = 1$, $m$ is a unit, and $M = (m)_R = R$, but this contradicts $M \subsetneq R$. We conclude $P = M$ is a maximal ideal. □

**Definition 10.2.**
Let $R$ be an integral domain.

**(1)** An element $r \in R \setminus \{0_R, \text{units}\}$ (nonzero and non unit) is called *irreducible* if

$$r = ab \quad \Longrightarrow \quad a \in R^\times \quad \text{or} \quad b \in R^\times.$$

**(2)** An element $p \in R \setminus \{0_R, \text{units}\}$ (nonzero and non unit) is called *prime* if

$$p/ab \quad \Longrightarrow \quad p/a \quad \text{or} \quad p/b.$$

**(3)** Two elements $a, b \in R$ are called *associate*; we write $a \sim b$, if

$$a = bu \qquad \text{where} \qquad u \in R^\times.$$

**Example 10.3.**
**(1)** A prime number $p \in \mathbb{Z}$ is both irreducible and prime in the above sense.

**(2)** An irreducible polynomial $P(X) \in \mathbb{Q}[X]$ is both irreducible and prime in the above sense.

**(3)** Let $x = a + ib \in \mathbb{Z}[i]$. Then $N(x) = a^2 + b^2 = 1$ implies either $a^2 = 0$, $b^2 = 1$, hence $x \in \{i, -i\}$, or $b^2 = 0$, $a^2 = 1$, hence $x = \{-1, 1\}$. In particular, $N(x) = 1 \Longrightarrow x \in (\mathbb{Z}[i])^\times$.

The element $1+i \in \mathbb{Z}[i]$ is irreducible: if $1+i = xy$ with $x, y \in \mathbb{Z}[i]$ then $N(1+i) = 1^2+1^2 = 2 = N(x)N(y)$ which implies $N(x) = 1$ or $N(y) = 1$ hence either $x$ is a unit or $y$ is a unit.

**(4)**

- $\mathbb{Z}^\times = \{-1, 1\}$ and $m, n \in \mathbb{Z}$ are associate if and only if $m = n$ or $m = -n$.
- In $\mathbb{Z}[i]$: $(2-i) \sim 1+2i$ since $(1+2i) = i(2-i)$ and $i \in \mathbb{Z}[i]^\times$.
- $f, g \in \mathbb{Q}[X]$ are associate if and only if $f(X) = \alpha g(X)$ with $\alpha \in \mathbb{Q}^\times$.

**Proposition 10.4.**
*In an integral domain $R$ a prime element is irreducible.*

*Proof.* Let $p \in R$ be a prime element we show $p$ is irreducible. Assume $p = ab$. Then $p/a$ or $p/b$. Suppose $p/a$ hence $a = pr$ for some $r \in R$. Then $p = ab = prb \Longrightarrow p - prb = p(1-rb) = 0_R$. But $p \neq 0_R$, hence (since $R$ is integral) $1 - rb = 0_R \Longleftrightarrow rb = 1 \Longrightarrow b$ is a unit. □

**Remark 10.3.**
The converse of Proposition 10.4 doesn't hold in general.

For example in $\mathbb{Z}[\sqrt{-5}]$ the element 3 is irreducible but is not a prime. Indeed suppose $3 = xy$ with $x, y \in \mathbb{Z}[\sqrt{-5}]$. Then $N(3) = 3^2 = N(x)N(y)$ hence $N(x) = 1, 3$, or 9. If $N(x) = 9$ then $N(y = c+\sqrt{-5})d) = c^2+5d^2 = 1$ and $y = 1$, or $y = -1$ is a unit, $N(x = a+b\sqrt{-5})) = a^2+5b^2 = 3$ is not possible, and if $N(x) = 1$ then $x$ is a unit. Thus 3 is irreducible.

Now $3^2 = (2+\sqrt{-5})(2-\sqrt{-5}) \Longrightarrow 3/(2+\sqrt{-5})(2-\sqrt{-5})$ but 3 divides neither $2+\sqrt{-5}$ nor $2-\sqrt{-5}$. Indeed if say $3(a+b\sqrt{-5}) = 2+\sqrt{-5}$ then $N(3)N(a+b\sqrt{-5}) = N(2+\sqrt{-5})$ which implies; since $N(3) = N(2+\sqrt{-5}) = 9$, that $N(a+b\sqrt{-5}) = a^2+5b^2 = 1$ hence $a+b\sqrt{-5} = 1$ or $a+b\sqrt{-5} = -1$ which implies $3 = 2+\sqrt{-5}$ or $-3 = 2+\sqrt{-5}$, this is absurd. This show 3 is not prime in $\mathbb{Z}[\sqrt{-5}]$.

**Definition 10.3.**
A **unique factorisation domain (UFD)** is an integral domain $R$ in which every element $r \in R \setminus \{0_R, \text{units}\}$ (non zero and non unit) has the following properties: **(i)** $r = p_1 p_2 \dots p_n$ is a product of irreducible elements $p_1, p_2, \dots, p_n$.

**(ii)** The above factorisation is unique up to associates, if $r = q_1 q_2 \dots q_m$ is another factorisation of $r$ as product of irreducible elements $q_1, q_2, \dots, q_m$ then $n = m$ and after possibly renumbering the factors $p_i \sim q_i$; i.e. $p_i = u_i q_i$ with $u_i \in R^\times$, for all $1 \leq i \leq n$.

**Example 10.4.**
**(1)** $\mathbb{Z}$ is a UFD.

**(2)** $\mathbb{Z}[\sqrt{-5}]$ is NOT a UFD. Indeed in $\mathbb{Z}[\sqrt{-5}]$ we have

$$6 = 2 \times 3 = (1+\sqrt{-5})(1-\sqrt{-5})$$

and 2, 3, $1+\sqrt{-5}$, and $1-\sqrt{-5}$ are irreducible.

We show 2 is irreducible: if $2 = (a+b\sqrt{-5})(c+d\sqrt{-5})$ then $2^2 = N(2) = N(a+b\sqrt{-5})N(c+d\sqrt{-5})$ hence $N(a+b\sqrt{-5}) = a^2+5b^2 = 1, 2$ or 4. Further $N(a+b\sqrt{-5}) = a^2+5b^2 = 2$ is not possible. Hence either $N(a+b\sqrt{-5}) = a^2+5b^2 = 1$ in which case $a+b\sqrt{-5} = -1, 1$ and $a+b\sqrt{-5}$ is a unit, or $N(a+b\sqrt{-5}) = a^2+5b^2 = 4$ hence $N(c+d\sqrt{-5}) = c^2+5d^2 = 1$ and $c+d\sqrt{-5}$ is a unit. Similarly one shows that 3, $1+\sqrt{-5}$, and $1-\sqrt{-5}$ are irreducible.

We have $(\mathbb{Z}[\sqrt{-5}])^\times = \{1, -1\}$, and 2, $1+\sqrt{-5}$, and $1-\sqrt{-5}$ are not associate. Thus the factorisation of 6 as a product of irreducible elements in $\mathbb{Z}[\sqrt{-5}]$ is not unique.

**Proposition 10.5.**
*In a unique factorisation domain $R$ an element is prime if and only if it is irreducible.*

*Proof.* We only need to prove that an irreducible element is prime (see Proposition 10.4). Let $p \in R$ be an irreducible element and suppose $p/ab$, thus $ab = pr$ for some $r \in R$, we must show that $p/a$ or $p/b$. Write $a, b$ and $r$ as a product of irreducible elements $a = p_1 p_2 \dots p_n$, $b = q_1 q_2 \dots q_m$, and $r = r_1 r_2 \dots r_t$. Thus we have

$$(p_1 p_2 \dots p_n)(q_1 q_2 \dots q_m) = p(r_1 r_2 \dots r_t).$$

The unique factorisation property implies $\exists 1 \leq i \leq n$ such that $p \sim p_i$, or $\exists 1 \leq j \leq m$ such that $p \sim q_j$. Suppose $p \sim p_i \iff p_i = up, u$ unit $\implies a = p_1 \dots up \dots p_n \implies p/a$. If $p \sim q_j \iff q_j = vp, v$ unit $\implies b = q_1 \dots vp \dots q_m \implies p/b$. □

**Proposition 10.6.**
*Let $a, b$ be nonzero elements of a unique factorisation domain $R$ and suppose $a = p_1^{e_1} p_2^{e_2} \dots p_n^{e_n}$, $b = p_1^{f_1} p_2^{f_2} \dots p_n^{f_n}$ are prime factorisations for $a$ and $b$, the primes $p_1, p_2, \dots, p_n$ are distinct, and the exponents $e_i, f_i$ are $\geq 0$. Then*

$$d = p_1^{\min(e_1, f_1)} p_2^{\min(e_2, f_2)} \dots p_n^{\min(e_n, f_n)}$$

*(where $d = 1$ if all exponents $\min(e_i, f_i)$, for $1 \leq i \leq n$, are $0$) is a* gcd *of $a$ and $b$.*

*Proof.* Since the exponents of each of the primes occurring in the factorisation of $d$ are no larger than the exponents occuring in the factorisation of both $a$ and $b$, $d$ is a divisor of both $a$ and $b$. Let $c \in R$, $c/a$, $c/b$, and $c = q_1^{r_1} q_2^{r_2} \dots q_m^{r_m}$ the prime factorisation of $c$. A prime $q_j$ dividing $c$ divides $a$ and $b$ hence equals $p_i$ for some $1 \leq i \leq n$, up to associates. Further in this case $q_j^{r_j}/p_i^{e_i}/a$, and $q_j^{r_j}/p_i^{f_i}/b$, which implies $q_j^{r_j}/p_i^{\min(e_i, f_i)}/d$. This shows $c/d$ hence $d = \gcd(a, b)$. □

We state without proof the following result.

**Theorem 10.7.**
*A PID is a UFD. In particular an ED (which is a PID by Theorem 10.1) is a UFD.*

## Summary

In summary we have the following strict containments of the various types of rings we encountered in this section

$$\Big\{(ED)\Big\} \subsetneq \Big\{(PID)\Big\} \subsetneq \Big\{(UFD)\Big\}.$$

**Example 10.5.**
**(1)** $\mathbb{Z}$ is a UFD.

**(2)** $\mathbb{Q}[X]$, and more generally $F[X]$ where $F$ is a field, is a UFD since it is an ED.

**(3)** $\mathbb{Z}[i]$ is a UFD since it is an ED.

**(4)** $R = \left\{ a + b(\frac{1+\sqrt{-19}}{2}) : a, b \in \mathbb{Z} \right\}$ is a PID but is NOT an ED.

**(5)** one can show that $\mathbb{Z}[X]$ is a UFD which is not a PID.